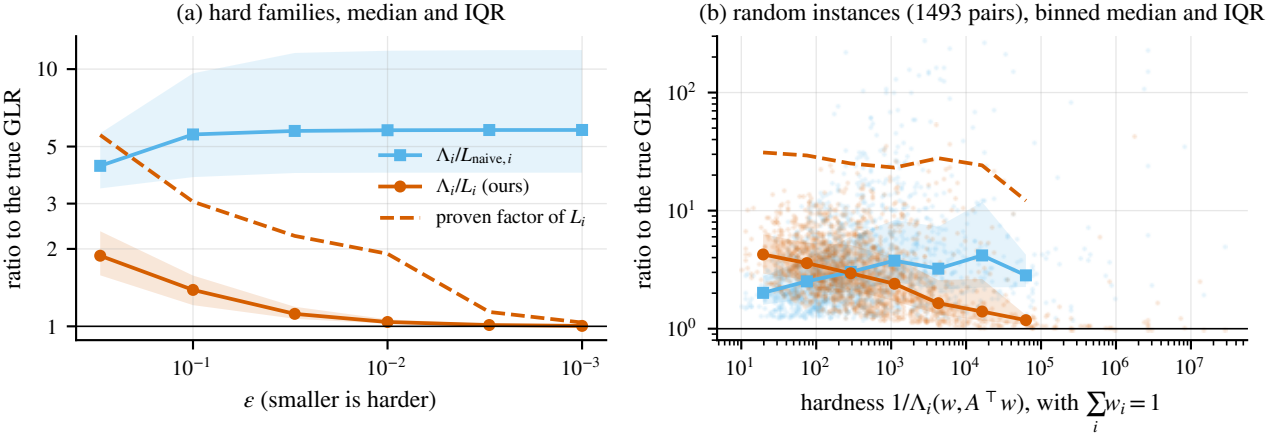
(a) hard families, median and IQR
ratio to the true GLR
10
5
3
2
1
$\Lambda_i/L_{\text{naive},i}$
Λ_i/L_i (ours)
proven factor of L_i
10^{-1}
10^{-2}
10^{-3}
ε (smaller is harder)
(b) random instances (1493 pairs), binned median and IQR
ratio to the true GLR
10^2
10^1
10^0
10^1
10^2
10^3
10^4
10^5
10^6
10^7
hardness $1/\Lambda_i(w, A^\top w)$, with $\sum_i w_i = 1$